



Digital Twins of Artifacts

Carter Laubach & Iris Angello · Advisor: Karen Braun
In collaboration with the RIT Cary Graphic Arts Collection

INTRODUCTION

Preserving fragile history as interactive replicas

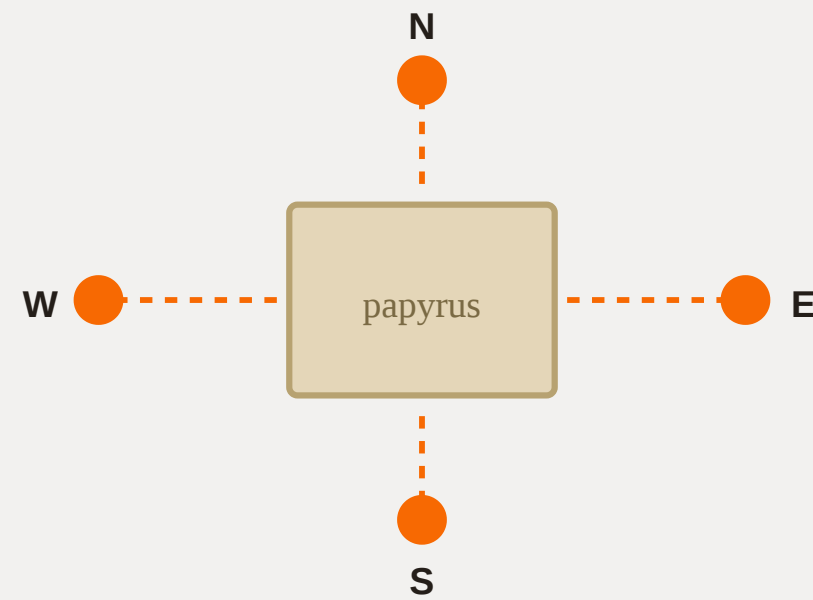
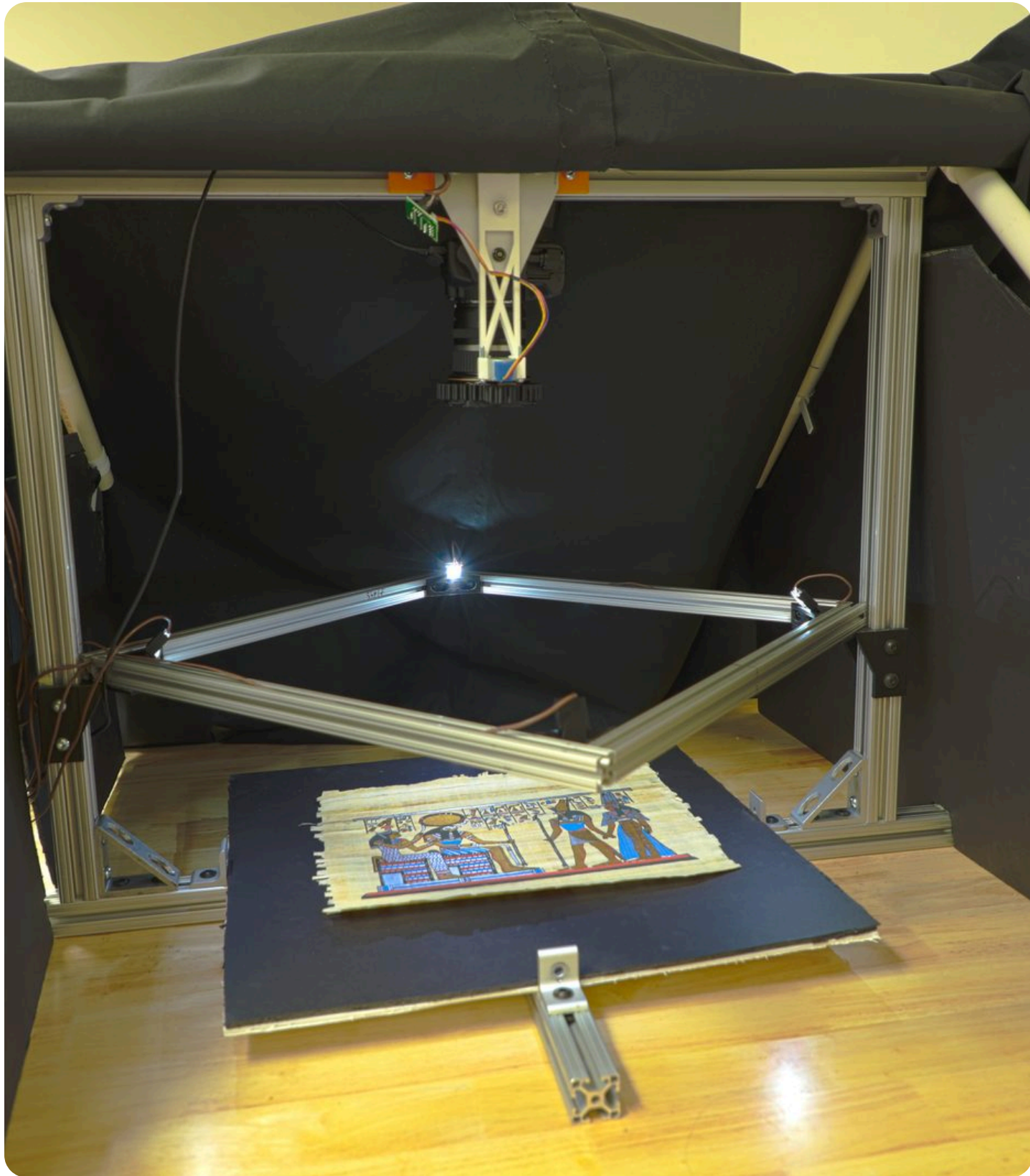
Have you ever stood in a museum and wished you could pick up the artifact behind the glass — turn it over, tilt it to the light, and look closer?

Papyrus manuscripts and cuneiform tablets are fragile, rare, and kept out of reach of the people who most want to study them. **Digital Twins of Artifacts** answers this by building accurate, web-viewable replicas that anyone can examine, teach with, and display — without ever handling the original.

SYSTEM A · PAPYRUS & FLAT MANUSCRIPTS

2D Photometric-Stereo Rig

A Canon DSLR paired with an Arduino-driven directional lighting array photographs the manuscript once per light direction. Instead of reconstructing geometry, it recovers how the surface *responds to light* — revealing texture, ink, and fine relief.



Four directional lights (N/E/S/W) at $\sim 37^\circ$. A rotating polarizer captures **cross-polarized** (clean geometry) and **co-polarized** (reveals ink) pairs.

APPROACH

Two purpose-built imaging systems

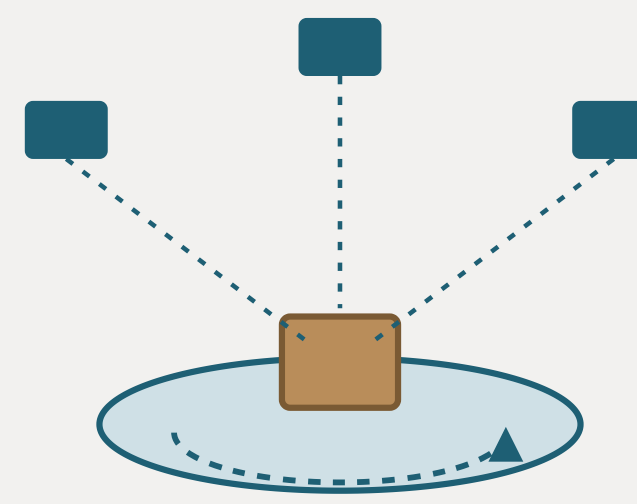
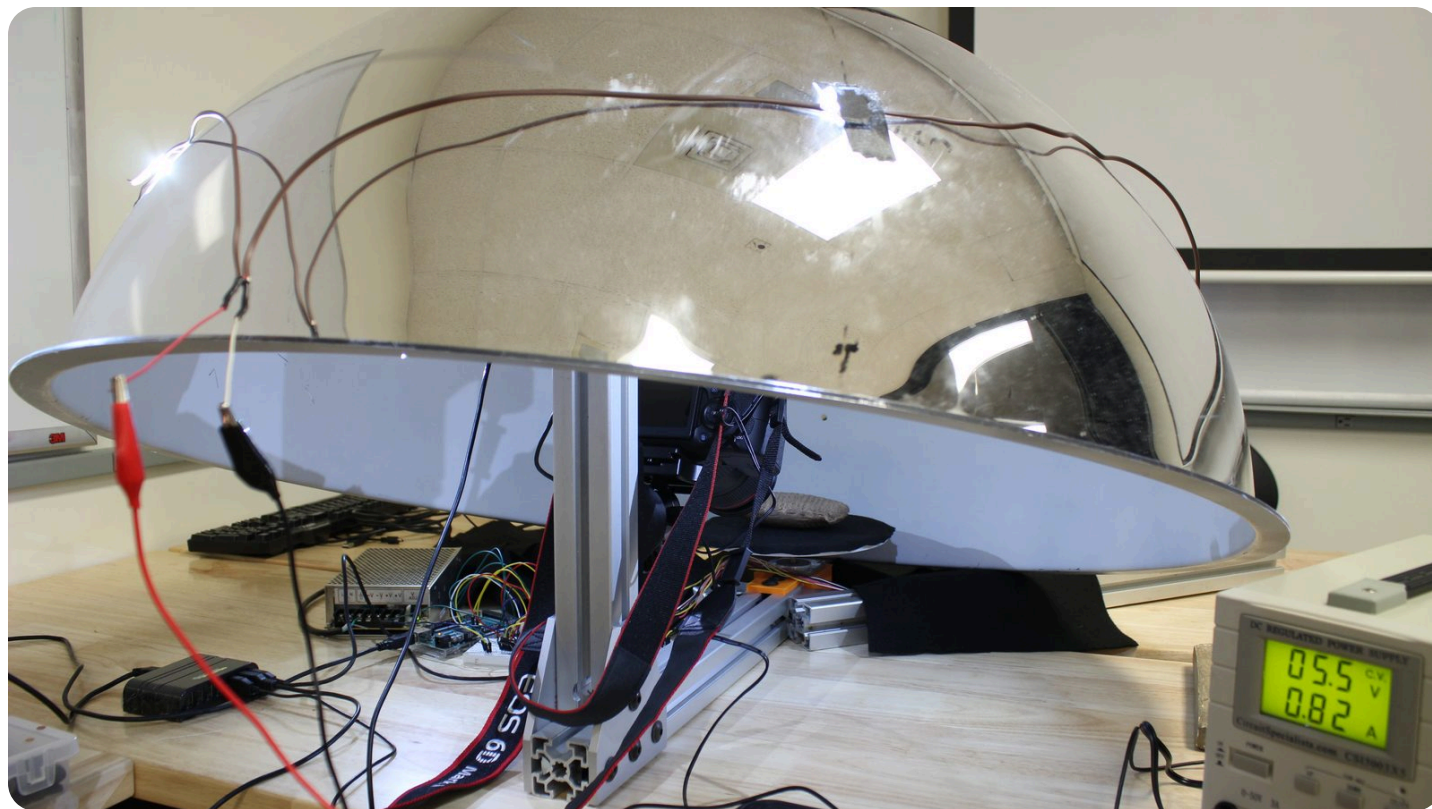
Each class of artifact is paired with a rig designed for it. Which one an object uses depends on a single question: **does its shape carry meaning in three dimensions?**

- **Flat, detail-rich** objects (papyrus, parchment) → a 2D photometric-stereo rig that recovers how the surface responds to light.
- **Truly three-dimensional** objects (cuneiform tablets, seals) → a 3D photogrammetry rig that reconstructs real geometry.

SYSTEM B · CUNEIFORM TABLETS & OBJECTS

3D Photogrammetry Rig

A multi-camera rig on an Arduino-driven turntable photographs the artifact from many angles. GPU-accelerated structure-from-motion and dense stereo build a point cloud and, ultimately, a watertight 3D mesh — a true replica of the object's form.



Multiple cameras shoot the object as the **turntable rotates**. Both sides are captured, then merged into one model.

ACCESS

An interactive web viewer

Every finished twin lives in a static three.js gallery that runs in any browser — rotate each artifact, adjust the lighting, and swap the backdrop to study the surface.

- **Hand-tracked magnifier** — a webcam tracks the visitor's fingertip and a magnifying lens follows it across the artifact, like a museum exhibit.
- **Adjustable relighting** reveals ink and relief a fixed view would hide.
- **Runs anywhere** — a static site, shared with a single link.

REFERENCE

Technical specifications

Camera	2D + 3D	Canon DSLRs, tethered. 2D: RAW → 16-bit TIFF ($\sim 5208 \times 3476$ px); 3D: JPG
Lighting	2D	4 directional lights (N/E/S/W) at $\theta \approx 37^\circ$; rotating polarizer
Polarization	2D	cross-pol (geometry) + co-pol (ink) capture pairs
Reconstruction	3D	CUDA COLMAP (SfM + MVS) → FPFH/RANSAC + ICP → Open3D Poisson mesh
Rendering	2D + 3D	three.js material baked to GLB; viewer w/ MediaPipe hand tracking

CONCLUSION & FUTURE WORK

A repeatable path to digital preservation

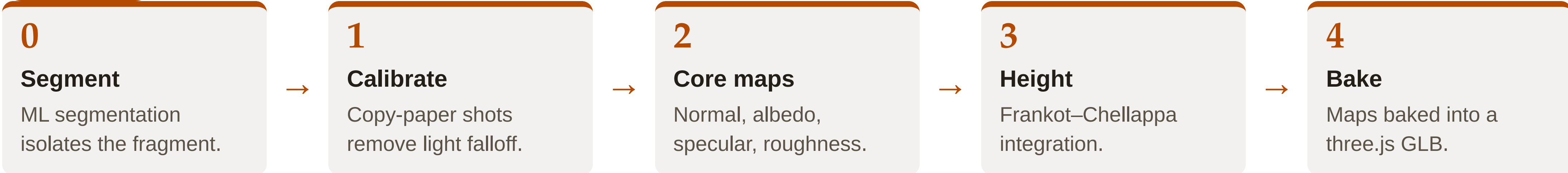
Two complementary rigs and their automated pipelines turn fragile originals into faithful, shareable digital twins — a workflow that repeats across artifact types.

- Expand the collection of scanned artifacts with the Cary Collection.
- Refine capture speed and automation for both rigs.

METHODS

From raw captures to a digital twin

PAPYRUS Photometric-stereo pipeline (Python)



TABLETS Photogrammetry pipeline (COLMAP / Open3D)



RESULTS

Outputs



Papyrus digital twin. A manuscript rebuilt from its polarized-light surface maps (normal, albedo, specular, height) and rendered in the web viewer.



Cuneiform-tablet twin. Both faces of the tablet reconstructed by photogrammetry and merged into one watertight, textured 3D model.



Hand-tracked magnifier. In the web viewer a webcam follows the visitor's fingertip, and a magnifying lens tracks across the artifact — a museum-style interactive.

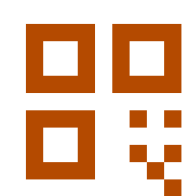
ACKNOWLEDGEMENTS

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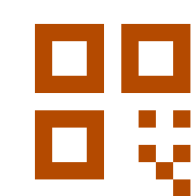
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Explore the digital twins

Contact: cjl6825@rit.edu · isa4049@rit.edu · Artifact questions: RIT Cary Graphic Arts Collection · Rochester Institute of Technology



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